Ahmedabad University

CSE623 Machine Learning

Weekly Report 4

Feature-Based Global Wheat Full Semantic Segmentation using Classical Machine Learning and Contour Analysis

Submitted to faculty: Prof. Mehul Raval

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Aim:

To develop a Feature Engineering Pipeline that extracts color, texture, and shape descriptors for pixel-wise classification in wheat head detection.

Introduction:

This report is about our progress on the Feature-Based Global Wheat Full Semantic Segmentation project.

Our main goal is to help precision farming and crop monitoring by identifying wheat plant parts like spikes, stems, leaves and background at the pixel level. This week our team worked on Phase 1: creating a Feature Engineering Pipeline. We extracted features like color, texture and shape from images to turn raw pixels into useful data. This helps us prepare our data for the phase, where we will classify the wheat plant parts. The Feature-Based Global Wheat Full Semantic Segmentation project will help us classify wheat plant parts accurately.

Work Completed:

This week, we tried to build the Feature Engineering Pipeline, which is the core module for extracting the descriptors needed for our Classical Machine Learning Model.

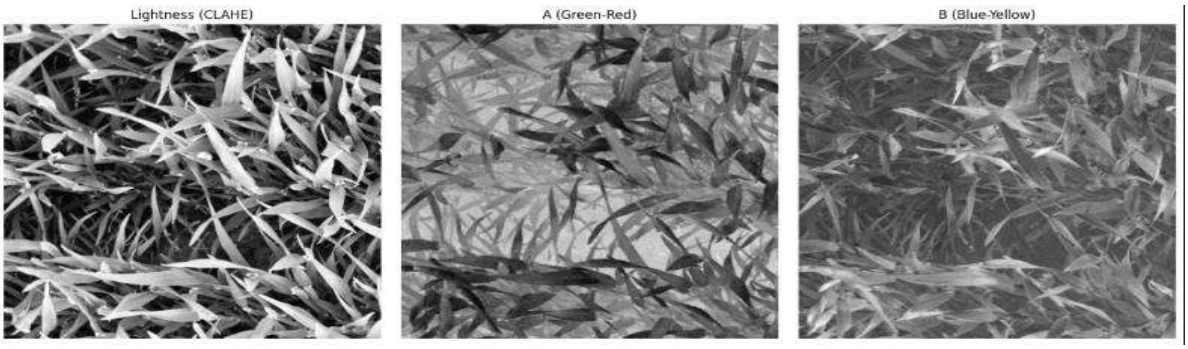
1. Color Descriptors:

Method Implemented: Conversion from RGB to LAB Color Space (utilizing the A and B channels).

Justification:

Raw RGB images get highly affected by the shadows, lighting, and based on the camera setup which can be confusing for a classifier. So when we change the image to LAB color space, we can separate the light that's actually on the object from the real color of the object. The light is in the L-channel. The color is in the A-channel for green and red and the B-channel for blue and yellow.

We take the A and B channels just as they are. Put them into the model. This way a green leaf that is in sunlight and a green leaf that is, in a dark shadow will be seen as having the same color, which makes the model work a lot better.

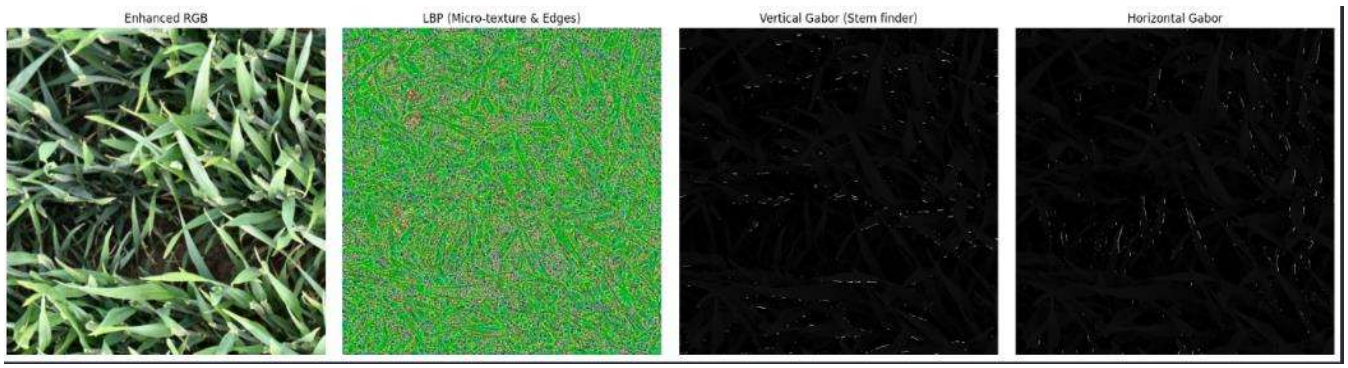


2. Texture Descriptors:

Method Implemented: Local Binary Patterns (LBP) and Gabor Filter Banks applied strictly to the structural L-channel.

Justification: We used LBP because it performs well to identify the details of textures. It assigns high mathematical values to the bumpy, grainy surface of wheat heads compared to the smooth surface of leaves.

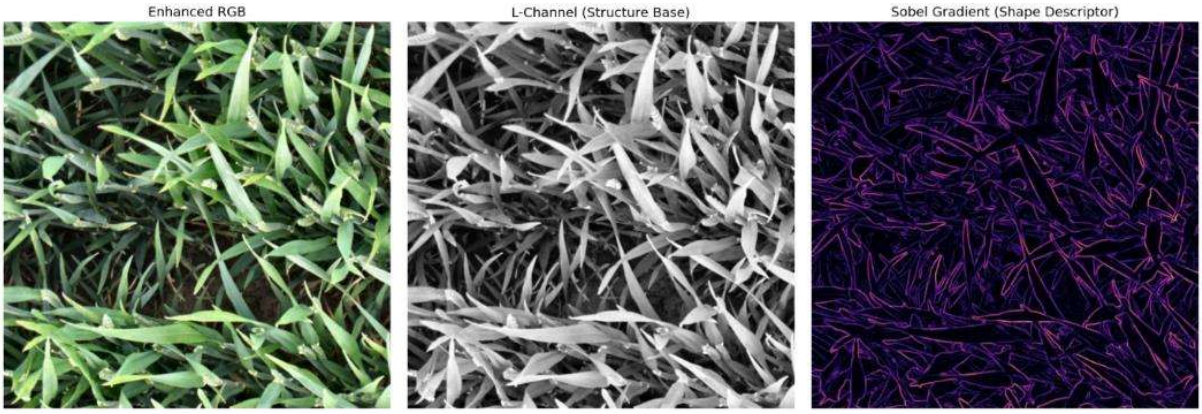
We also used Gabor filters as mentioned in the problem description. These filters evaluate texture frequency at specific angles. Because wheat stems are biologically distinct vertical tubes, a vertical Gabor filter acts as an ideal "stem detector," while horizontal or diagonal filters help capture leaf orientations.



3. Shape Descriptors:

Justification:

When we look at a pixel we cannot justify the actual shape, like how round or square it is. We use the Sobal filter which computes the mathematical difference between neighbouring pixels. And makes it easy to identify and differentiate the actual shape of different shape present in the image.



Next steps and goals:

- With the feature engineering pipeline complete our focus will now shift to Phase 2, The Classical ML Segmentation Model. The main goals are:

- Data Preparation: We need to flatten the 2D feature maps we got, like Color, LBP, Gabor and Sobel into 1D arrays. Then we stack them into one Pandas DataFrame. This Data Frame should match the datasets ground truth masks perfectly.

- Model Training: Next we will train a -neural classifier. A Random Forest classifier seems like a choice because it can handle huge pixel-level datasets well. We want this classifier to tell wheat spikes, leaves, stems and background soil apart using our hand-crafted features.

- Model Evaluation: After that we will check the models accuracy and feature importance. This will help us see which features, like texture or color are driving the models decisions. We want to understand what the model is using to make its choices.

Conclusion:

This week we got the feature engineering pipeline to work. This gives a base, for the work we are doing to separate into groups. We used ways of looking at pictures, and extracted the impotent features from the image. Now that we are starting the part of the project we can use the good information we found to help our classifier to perform well in figuring out what is going on with the wheat at the level of each tiny piece of the picture.